

12.4 Cross Products

Calculus III

0.1 Cross Product

Given two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$, the **cross product** $\mathbf{u} \times \mathbf{v}$ is a vector with magnitude

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta \quad (1)$$

where $0 \leq \theta \leq \pi$ is the angle between $\mathbf{u}$ and $\mathbf{v}$. The direction of $\mathbf{u} \times \mathbf{v}$ is given by the **right-hand rule**: When you put the vectors tail to tail and let the fingers of your right hand curl from $\mathbf{u}$ to the direction of $\mathbf{v}$, the direction of your thumb, orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, is the direction of $\mathbf{u} \times \mathbf{v}$. When $\mathbf{u} \times \mathbf{v} = \mathbf{0}$, the direction of $\mathbf{u} \times \mathbf{v}$ is undefined.

0.1.1 Geometry of the Cross Product

Let $\mathbf{u}$ and $\mathbf{v}$ be two nonzero vectors in $\mathbb{R}^3$.

1. The vectors $\mathbf{u}$ and $\mathbf{v}$ are parallel ($\theta = 0$ or $\theta = \pi$) if and only if $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.
2. If $\mathbf{u}$ and $\mathbf{v}$ are two sides of a parallelogram, then the area of the parallelogram is

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}| |\mathbf{v}| \sin \theta \quad (2)$$

0.1.2 Properties of the Cross Product

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be nonzero vectors in $\mathbb{R}^3$, and let a and b be scalars.

$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$	Anticommutative property	(3)
$(a\mathbf{u}) \times (b\mathbf{v}) = ab(\mathbf{u} \times \mathbf{v})$	Associative property	(4)
$\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) + (\mathbf{u} \times \mathbf{w})$	Distributive property	(5)
$(\mathbf{u} + \mathbf{v}) \times \mathbf{w} = (\mathbf{u} \times \mathbf{w}) + (\mathbf{v} \times \mathbf{w})$	Distributive property	(6)

0.1.3 Cross Products of Coordinate Unit Vectors

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60125 [tdplot_main_coords, cube/.style = verythick, black, grid/.style = verythin, gray, axis/.style =
      ->, blue, thick]
      in -1.5,0,...,3.5 in -1.5,0,...,3.5 [grid] (-1.5,) - (3.5); [grid] (-1.5,) - (3.5);
[axis] (0,0,0) - (5,0,0) node[anchor=west]x =; [axis] (0,0,0) - (0,5,0) node[anchor=west]y =; [axis] (0,0,0) -
      (0,0,5) node[anchor=west]z =;
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$$\mathbf{i} \times \mathbf{j} = \mathbf{k} \quad (7)$$

$$\mathbf{j} \times \mathbf{k} = \mathbf{i} \quad (8)$$

$$\mathbf{k} \times \mathbf{i} = \mathbf{j} \quad (9)$$

$$\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = \mathbf{0} \quad (10)$$

0.1.4 Evaluating the Cross Product

Let $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$. Then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k} \quad (11)$$